

# Kaiwen (Kevin) He

Cambridge, MA | khe01@mit.edu | github.com/kevin-he-01 | Website: <https://kevin-he-01.github.io/>

## Research Interests

---

**Secure multi-party computation (MPC):** joint computation on private data while maintaining individual privacy. I am especially interested in the application of homomorphic secret sharing (HSS) techniques to MPC.

## Education

---

|                                              |                     |
|----------------------------------------------|---------------------|
| <b>Massachusetts Institute of Technology</b> | Cambridge, MA       |
| • Ph.D. candidate in Computer Science        | Sep 2023 – Current  |
| • M.S. in Computer Science                   | Sep 2023 – Sep 2025 |

|                                           |                     |
|-------------------------------------------|---------------------|
| <b>University of California San Diego</b> | La Jolla, CA        |
| • B.S. in Computer Engineering            | Sep 2020 – Jun 2023 |

## Experience

---

|                                                                                                                           |                     |
|---------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Research Intern</b> , Aarhus University – Aarhus, Denmark                                                              | May 2026 – Aug 2026 |
| • Investigated security of MPC in practice: <a href="https://mpcinthewild.github.io/">https://mpcinthewild.github.io/</a> |                     |

|                                                 |                     |
|-------------------------------------------------|---------------------|
| <b>Teaching Assistant</b> , MIT – Cambridge, MA | Jan 2026 – May 2026 |
| • 6.5610, Applied Cryptography.                 |                     |

|                                                                                                         |                     |
|---------------------------------------------------------------------------------------------------------|---------------------|
| <b>Research Experiences for Undergraduates</b> , UC San Diego – La Jolla, CA                            | Jun 2022 – May 2023 |
| • Collected weekly data from $2^{32}$ or 4 billion hosts (the entire IP address space).                 |                     |
| • Designed a new open source ZGrab 2.0 module with 7829 lines of code to collect data from IPsec hosts. |                     |

## Academic Services

---

**External Reviewer**, CANS 2026

**External Reviewer**, Proceedings of the IEEE

## Talks

---

|                                                       |                 |
|-------------------------------------------------------|-----------------|
| <b>Aarhus University Crypto Summer Day</b>            | Aarhus, Denmark |
| <i>Dyad (ongoing work on multi-party computation)</i> | July 2026       |

|                                                                                   |                 |
|-----------------------------------------------------------------------------------|-----------------|
| <b>Aarhus University Crypto Seminar</b>                                           | Aarhus, Denmark |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | June 2026       |

|                                                                                   |                   |
|-----------------------------------------------------------------------------------|-------------------|
| <b>IEEE S&amp;P</b>                                                               | San Francisco, CA |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | May 2026          |

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
| <b>Northeastern University</b> , Crypto Reading Group                             | Boston, MA |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | March 2026 |

|                                                                                   |               |
|-----------------------------------------------------------------------------------|---------------|
| <b>Boston University</b> , BUsec Seminar                                          | Boston, MA    |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | February 2026 |

|                                                                                   |              |
|-----------------------------------------------------------------------------------|--------------|
| <b>Google</b>                                                                     | New York, NY |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | January 2026 |

|                                                                                   |              |
|-----------------------------------------------------------------------------------|--------------|
| <b>Berkeley Security Seminar</b>                                                  | Berkeley, CA |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | January 2026 |

|                                                                                   |              |
|-----------------------------------------------------------------------------------|--------------|
| <b>UCSD Security Lunch</b>                                                        | La Jolla, CA |
| <i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i> | January 2026 |

|                                            |               |
|--------------------------------------------|---------------|
| <b>MIT CIS Seminar</b> (with Lali Devadas) | Cambridge, MA |
|--------------------------------------------|---------------|

|                                                                      |                     |
|----------------------------------------------------------------------|---------------------|
| <i>Multi-Key Homomorphic Secret Sharing, From Theory To Practice</i> | December 2025       |
| <b>CSAW</b>                                                          | New York, NY        |
| <i>Passive SSH Key Compromise via Lattices</i>                       | November 2024       |
| <b>ACM CCS</b>                                                       | Copenhagen, Denmark |
| <i>Passive SSH Key Compromise via Lattices</i>                       | November 2023       |

## Awards and Honors

|                                                                                                                                                                             |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Recipient</b> , Qualcomm Innovation Fellowship                                                                                                                           | May 2026       |
| <i>Making Modern Cryptography Faster and More Energy-Efficient Through Hardware Acceleration of Modular Arithmetic</i>                                                      |                |
| <b>Most notable paper: technical impact</b> , CSAW Applied Research Competition                                                                                             | November 2024  |
| <i>Passive SSH Key Compromise via Lattices</i>                                                                                                                              |                |
| <b>Recipient</b> , Irwin Mark Jacobs and Joan Klein Jacobs Presidential Fellowship, MIT                                                                                     | September 2023 |
| <i>Offered to newly admitted Ph.D. students who have demonstrated exemplary academic and research achievements, and thus show great promise for future accomplishments.</i> |                |
| <b>Recipient</b> , SIM San Diego Scholarship, Society of Information Management (SIM) San Diego                                                                             | October 2022   |
| <i>Offered to nominated students only.</i>                                                                                                                                  |                |

## Publications

|                                                                                                                                                                                                   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>VROOM: Accelerating (Almost All) Number-Theoretic Cryptography Using Vectorization and the Residue Number System</b>                                                                           | August 2026   |
| Simon Langowski, <a href="#">Kaiwen He</a> , Srinivas Devadas                                                                                                                                     |               |
| <i>USENIX Security 2026</i>                                                                                                                                                                       |               |
| <b>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</b>                                                                                                                 | May 2026      |
| <a href="#">Kaiwen He</a> , Sacha Servan-Schreiber, Geoffroy Couteau, Srinivas Devadas                                                                                                            |               |
| <i>IEEE S&amp;P 2026</i>                                                                                                                                                                          |               |
| <b>Passive SSH Key Compromise via Lattices</b>                                                                                                                                                    | November 2023 |
| Keegan Ryan, <a href="#">Kaiwen He</a> , George Arnold Sullivan, Nadia Heninger                                                                                                                   |               |
| <i>ACM CCS 2023</i>                                                                                                                                                                               |               |
| <b>Critique of: “A Parallel Framework for Constraint-Based Bayesian Network Learning via Markov Blanket Discovery” by SCC Team From UC San Diego</b>                                              | October 2022  |
| Arunav Gupta, John Ge, John Li, Zihao Kong, <a href="#">Kaiwen He</a> , Matthew Mikhailov, Bryan Chin, Xiaochen Li, Max Apodaca, Paul Rodriguez, Mahidar Tatineni, Mary Thomas, and Santosh Bhatt |               |
| <i>IEEE TPDS 2022</i>                                                                                                                                                                             |               |

## Skills

**Programming Languages:** Python, JavaScript, Go, Java, Bash, C, C++, Rust, TypeScript, x86 and ARM Assembly, Kotlin.  
**Other:** Cryptography, Cryptanalysis, Technical Writing, Technical Presentation, Research.